



Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity

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Contents



- Existing Work (Mixture of Experts)
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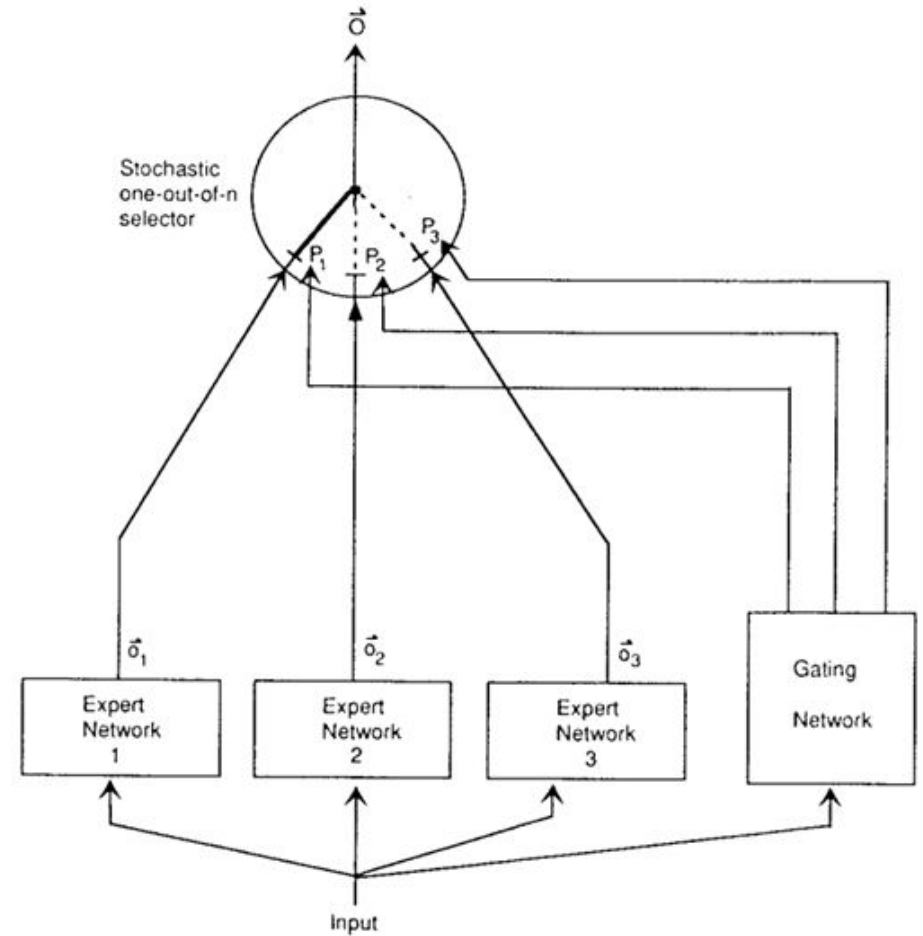
Mixture of Experts



Background of MoE (~30 years ago)



- Dates back at least 30 years to the work *Adaptive Mixtures of Local Experts*^[1].
- In early concepts, an expert was defined as an entire neural network and the MoE was similar to ensemble methods^[2].



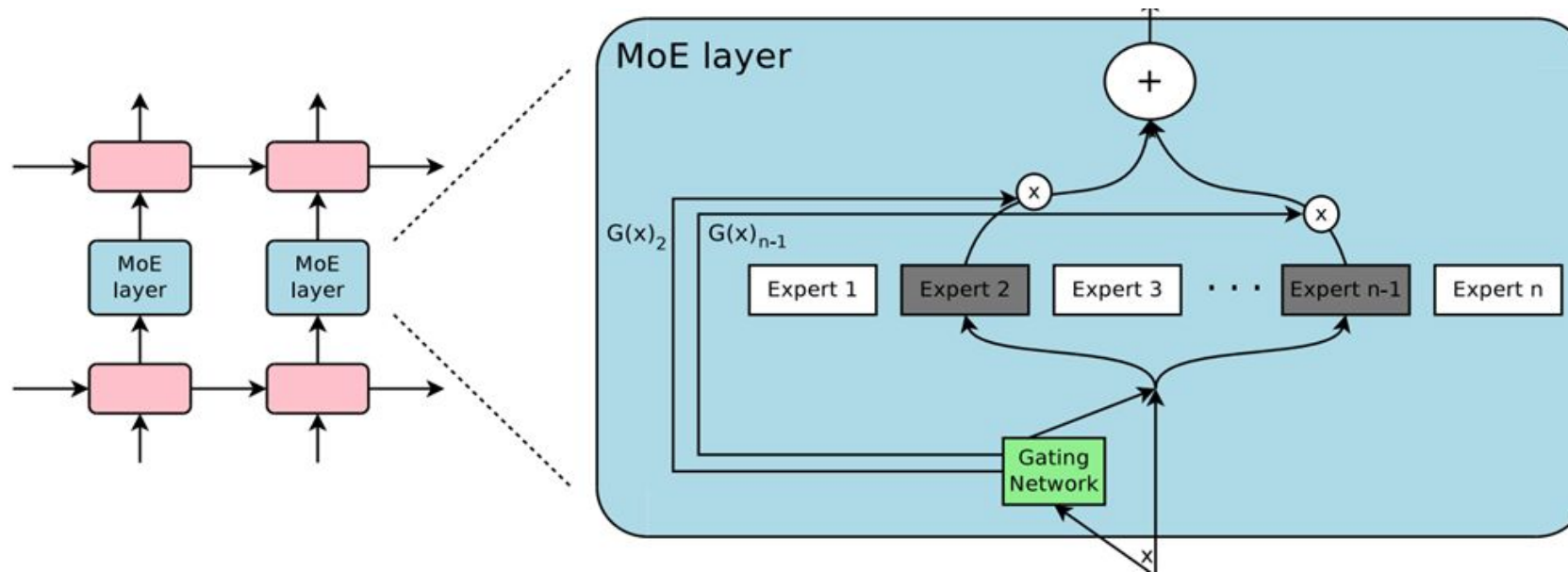
[1] R. A. Jacobs, M. I. Jordan, S. J. Nowlan and G. E. Hinton, "Adaptive Mixtures of Local Experts," , Neural Computation

[2] William Fedus et. al. A Review of Sparse Expert Models in Deep Learning

Background of MoE (RNN Era)^[3]



- Backbone: RNN
- MoE layer: multiple FFN experts
- Gating/routing network (also a FFN): assign tokens into different experts

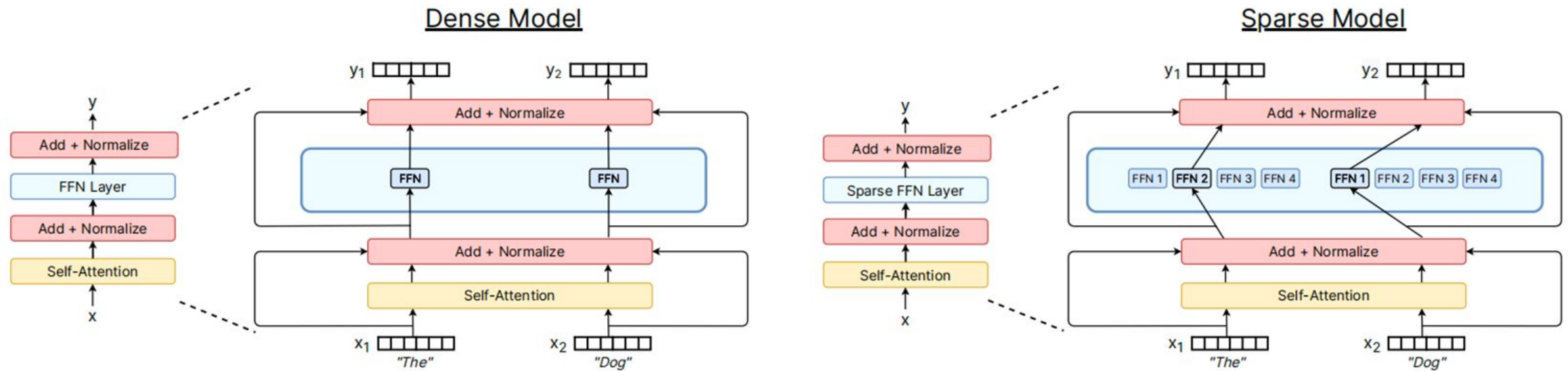


[3] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer

Background of MoE (Transformer Era)



- Replace the original FFN layers with sparse FFN ones (expert layers).



Motivation



Compared to dense models, MoE has two major advantages:

1. **More unique model parameters** can be included given similar computational budgets (FLOPs), which improves model performance empirically;

Model	Parameters	FLOPS	SuperGLUE (↑)
T5-Base	223M	124B	74.6
Switch-Base	7410M	124B	81.3

Motivation



2. **Functional specialization** among expert modules, which improves the model interpretability.

Expert	Top-5 preceding tokens
5	year, years, billion, millions, tonnes
9	electronic, local, public, national, outdoor
34	to, will, should it, may
42	two, 50, 1, 80, 000
62	work, started, involved, working, launched
72	is, was, be, been, were
74	going, go, come, back, return
101	B, T, W, H, k





Switch Transformer



Switch Routing: Rethinking Mixture-of-Experts.



Mixture of Experts

Route to k experts, where $k > 1$

$$h(x) = W_r \cdot x$$

$$p_i(x) = \frac{e^{h(x)_i}}{\sum_j^N e^{h(x)_j}}.$$

$$y = \sum_{i \in \mathcal{T}} p_i(x) E_i(x).$$

Switch Transformer

Only route to a single expert ($k = 1$)

Benefits:

1. Router computation is reduced.
2. Batch size of each expert can be halved.
3. Communication costs are reduced.

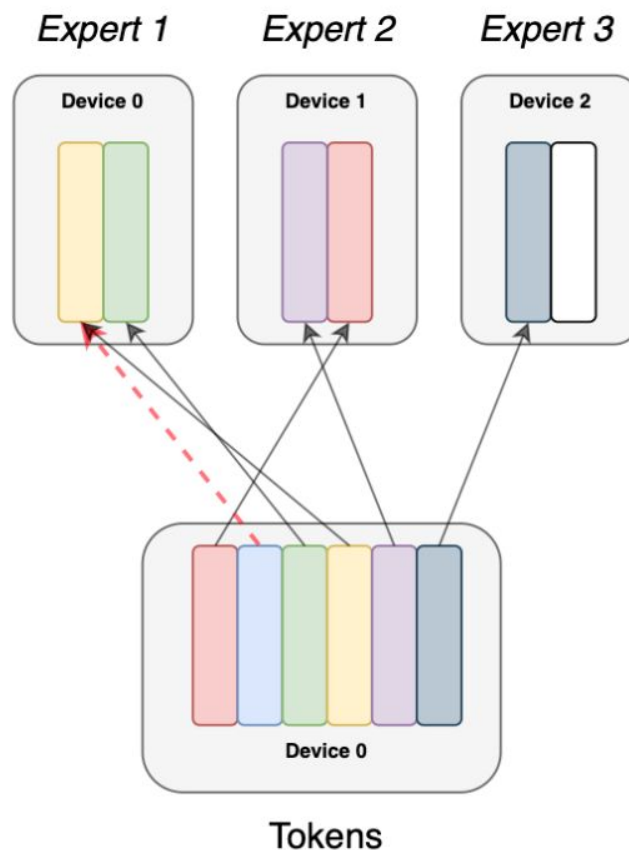
Expert Capacity



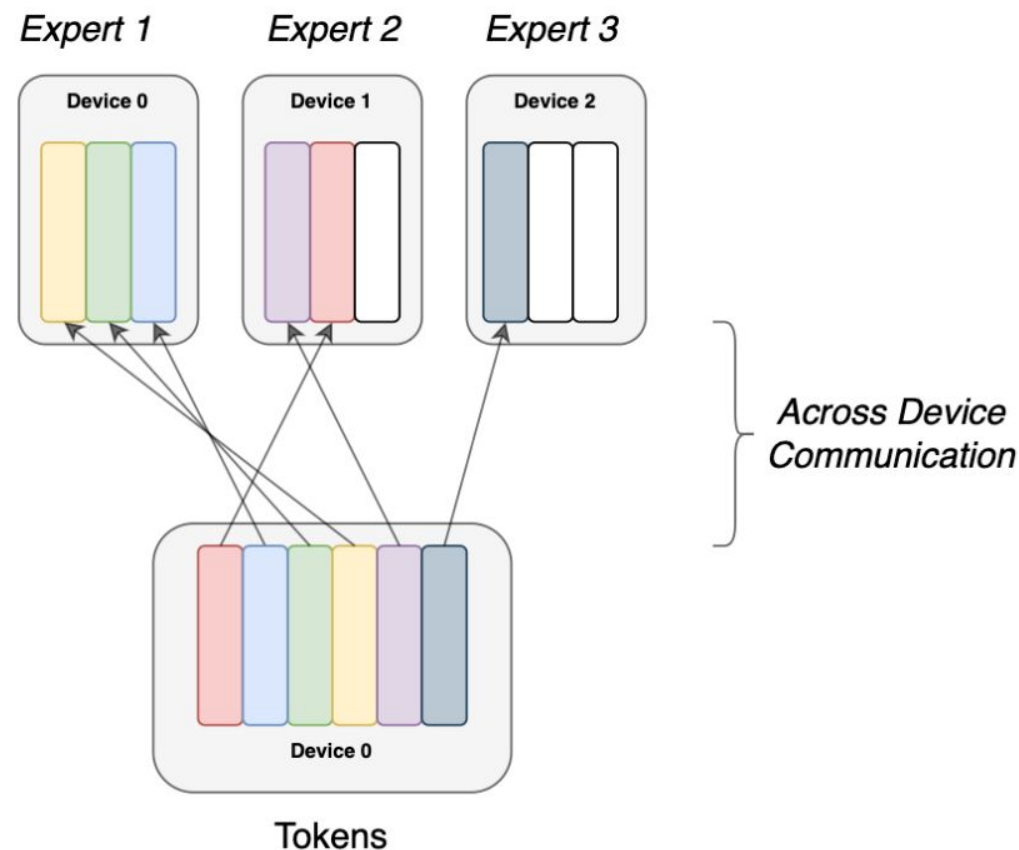
Terminology

- **Experts:** Split across devices, each having their own unique parameters. Perform standard feed-forward computation.
- **Expert Capacity:** Batch size of each expert. Calculated as $(\text{tokens_per_batch} / \text{num_experts}) * \text{capacity_factor}$
- **Capacity Factor:** Used when calculating expert capacity. Expert capacity allows more buffer to help mitigate token overflow during routing.

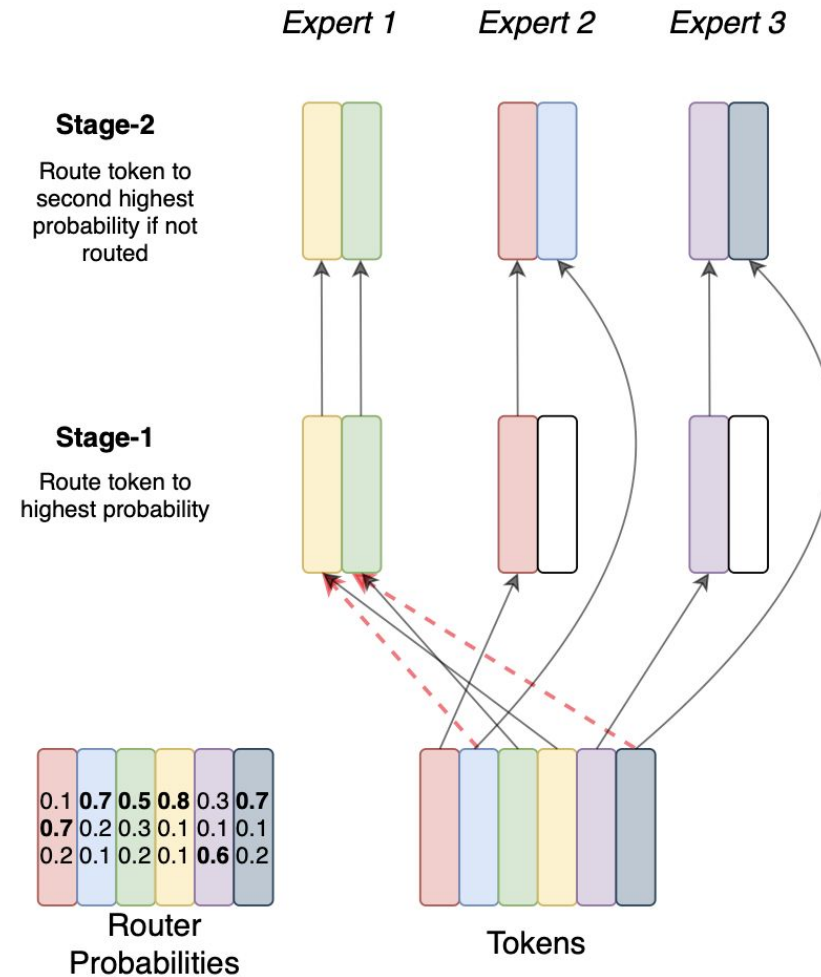
(Capacity Factor: 1.0)



(Capacity Factor: 1.5)



Preventing Token Dropping with No-Token-Left-Behind



A Differentiable Load Balancing Loss



$$\text{loss} = \alpha \cdot N \cdot \sum_{i=1}^N f_i \cdot P_i \quad (4)$$

where f_i is the fraction of tokens dispatched to expert i ,

$$f_i = \frac{1}{T} \sum_{x \in \mathcal{B}} \mathbb{1}\{\text{argmax } p(x) = i\} \quad (5)$$

and P_i is the fraction of the router probability allocated for expert i ,²

$$P_i = \frac{1}{T} \sum_{x \in \mathcal{B}} p_i(x). \quad (6)$$



Improved Training and Fine-Tuning Techniques

Issues with Mixture of Experts (MoE)

- Model complexity
- High communication costs
- Training instabilities

To improve training and fine-tuning of sparse models, Switch Transformers used...

1. Selective precision with large sparse models.
2. Smaller parameter initialization for stability.
3. Regularizing large sparse models.



1. Selective precision with large sparse models

- Model instability hinders the ability to train using efficient bfloat16 precision, so float32 precision was used for MoE.
- **Solution:** Selectively cast to float32 precision
 - Calculations within the router function are done in float32 precision
 - Resulting dispatch and combine tensors are recast to bfloat16

Model (precision)	Quality (Neg. Log Perp.) (↑)	Speed (Examples/sec) (↑)
Switch-Base (float32)	-1.718	1160
Switch-Base (bfloat16)	-3.780 [<i>diverged</i>]	1390
Switch-Base (Selective precision)	-1.716	1390

2. Smaller parameter initialization for stability



They found that sparse expert models are sensitive to initialization scale. **They reduced the default Transformer initialization scale $s = 1.0$ by a factor of 10.**

Model (Initialization scale)	Average Quality (Neg. Log Perp.)	Std. Dev. of Quality (Neg. Log Perp.)
Switch-Base (0.1x-init)	-2.72	0.01
Switch-Base (1.0x-init)	-3.60	0.68

3. Regularizing large sparse models



Consider pretraining and then fine-tuning on smaller downstream tasks. Switch transformers have high parameter counts and are prone to overfitting.

Model (dropout)	GLUE	CNNDM	SQuAD	SuperGLUE
T5-Base (d=0.1)	82.9	19.6	83.5	72.4
Switch-Base (d=0.1)	84.7	19.1	83.7	73.0
Switch-Base (d=0.2)	84.4	19.2	83.9	73.2
Switch-Base (d=0.3)	83.9	19.6	83.4	70.7
Switch-Base (d=0.1, ed=0.4)	85.2	19.6	83.7	73.0



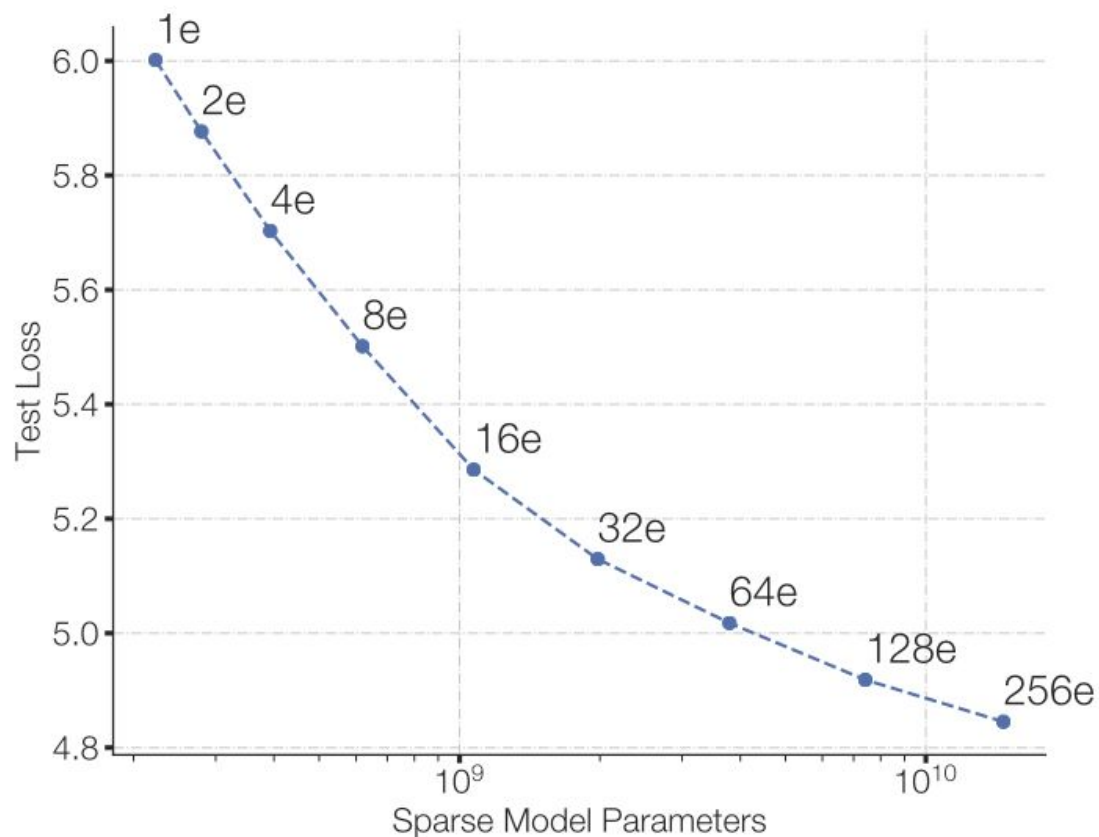
Scaling Properties



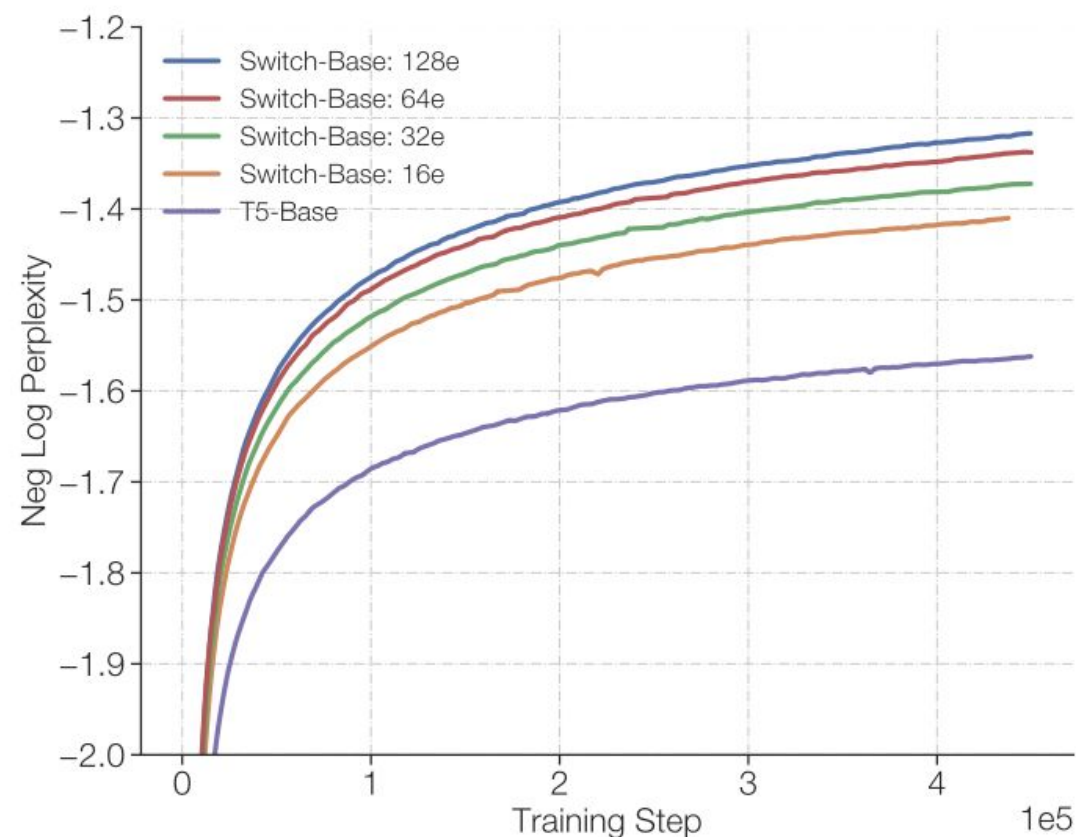
Scaling Results on a Step-Basis



Number of experts



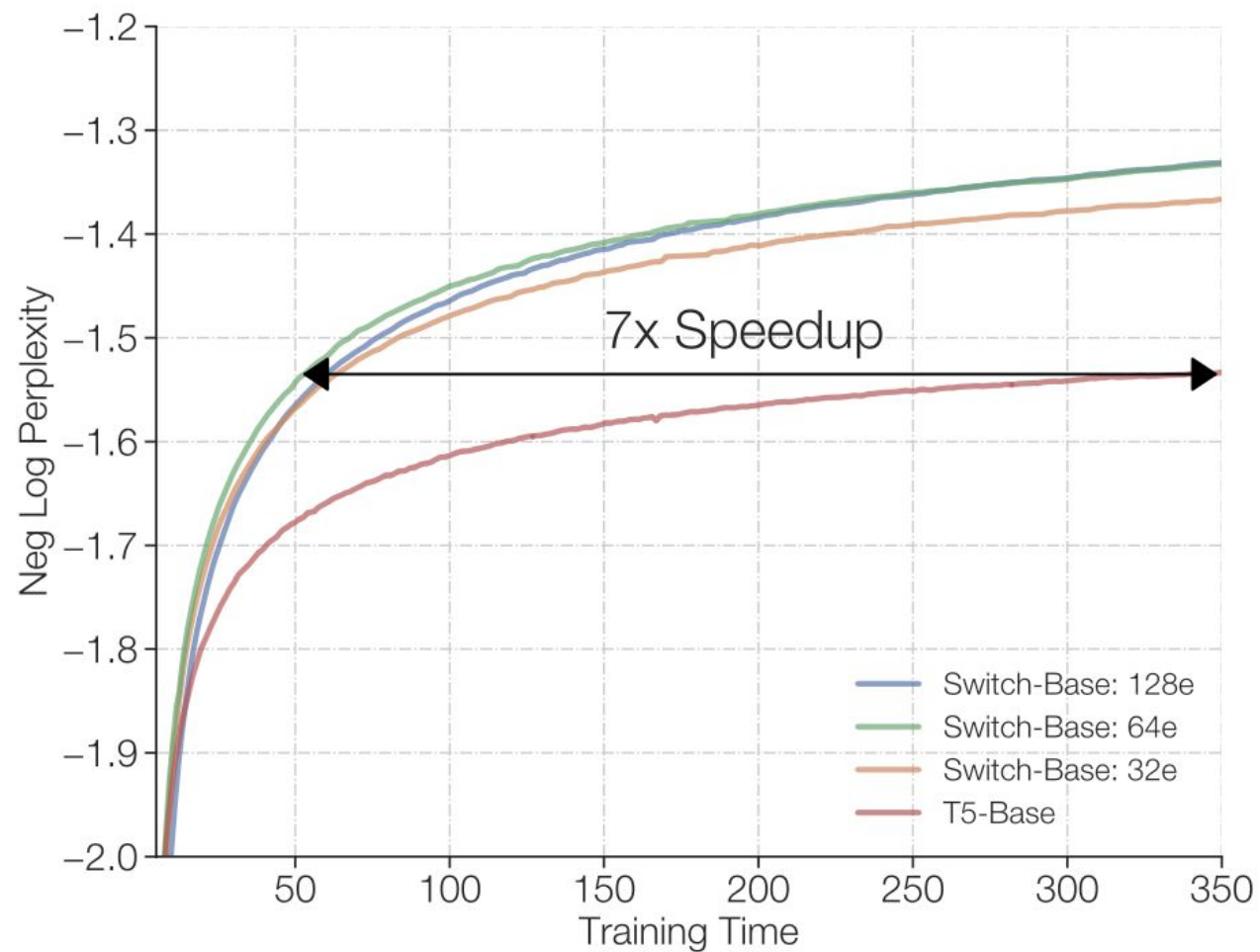
Training steps



Scaling Results on a Time-Basis



Wall-clock time





Fine-Tuning



Fine-tuning results



Model	GLUE	SQuAD	SuperGLUE	Winogrande (XL)
T5-Base	84.3	85.5	75.1	66.6
Switch-Base	86.7	87.2	79.5	73.3
T5-Large	87.8	88.1	82.7	79.1
Switch-Large	88.5	88.6	84.7	83.0

Model	XSum	ANLI (R3)	ARC Easy	ARC Chal.
T5-Base	18.7	51.8	56.7	35.5
Switch-Base	20.3	54.0	61.3	32.8
T5-Large	20.9	56.6	68.8	35.5
Switch-Large	22.3	58.6	66.0	35.5

Model	CB Web QA	CB Natural QA	CB Trivia QA
T5-Base	26.6	25.8	24.5
Switch-Base	27.4	26.8	30.7
T5-Large	27.7	27.6	29.5
Switch-Large	31.3	29.5	36.9